Vrinda Store: Why Revenue Fell 20% in 2022

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Skills demonstrated: SQL (CTEs, window functions, conditional aggregation), business analysis, executive communication

Executive Summary

The question I was given

Vrinda Store's revenue declined from a March 2022 peak of Rs. 1.82M to a November low of Rs. 1.46M, a 20% drop in eight months. The goal of this analysis was to understand what drove the decline and identify where the business should focus its recovery efforts.

What I found

The decline was concentrated rather than business-wide. Revenue from women customers fell by 22–30% across all age groups, compared with a 5–8% decline among men. Most of the revenue loss came from Amazon and Myntra and was concentrated in the Set and Kurta categories.
Average order value remained relatively stable, suggesting that the decline was driven primarily by fewer orders rather than lower transaction values.

What I recommend

Vrinda should prioritize re-engaging women customers on Myntra and Amazon, with an initial focus on the Set and Kurta categories.
If the business recovers 50% of the total revenue lost from women customers between March and November, the potential annual revenue recovery is approximately Rs. 2 M. This estimate should be treated as a scenario rather than a guaranteed outcome.

How I got there

I used six SQL queries to investigate the decline from several angles: monthly revenue trends, channel performance, category performance, customer demographics, and women's revenue performance by channel. Each stage narrowed the analysis from the overall decline to the segments where the largest losses occurred.

The Investigation: How I Found the Answer

A 20% revenue decline could have several causes. I investigated the data step by step, starting with the overall trend and then narrowing the analysis by channel, category, and customer segment.

Step 1: Establish the Trend

First, I analyzed monthly revenue, orders, and average order value across 2022 to understand how the decline developed.

What this tells us:

Revenue declined from March to November, while orders fell by approximately 22%. Average order value remained relatively stable, with a slight increase toward the end of the year. This suggests that the revenue decline was driven mainly by fewer orders rather than a significant change in order value.

Step 2: Identify the Channels with the Largest Revenue Declines

Next, I compared revenue by sales channel between March, the peak month, and November, the lowest month, to determine where the largest losses occurred.

What this tells us

Amazon and Myntra together account for Rs. 261K of the Rs. 363K monthly decline, or 72% of the total loss. Nalli had the largest percentage decline, but its absolute revenue loss was Rs. 35K, compared with Rs. 143K for Amazon. This shows why both percentage change and absolute impact need to be considered when prioritizing where to act.

Step 3: Identify the Categories Driving the Decline

I then compared category revenue between March and November to determine which product categories contributed most to the overall decline.

What this tells us

Set and kurta together account for approximately Rs. 354K of the Rs. 363K monthly revenue decline, or 97% of the total loss. The remaining categories had a much smaller impact. At this stage, the analysis narrowed the main decline to two product categories: Set and Kurta.

Step 4: Identify the Customer Segment Behind the Decline

After identifying the main channels and categories, I compared revenue by gender and age group to see whether the decline affected all customers equally.

What this tells us

Women customers recorded much larger revenue declines across every age group, falling by 22.8% to 29.8%. Revenue from men declined by only 5.1% to 7.7%. This was the clearest indication that the overall revenue decline was concentrated among women customers rather than affecting the customer base equally.

Step 5: Test Where the Women-Customer Decline Was Concentrated

I then compared March and November revenue for women customers by channel to determine whether the decline was concentrated in the same marketplaces identified earlier.

What this tells us

Myntra recorded the largest monthly revenue loss among women customers at approximately Rs.112K, followed by Amazon at Rs.89K. Together, the two channels account for approximately Rs.202K of the monthly decline among women customers. This confirms that the strongest opportunity for further investigation and recovery is among women customers on Myntra and Amazon.

Recommendations

The analysis identified where the largest revenue declines occurred. The following recommendations focus on the areas with the greatest potential impact.

Priority 1: Investigate and re-engage women customers on Myntra

Myntra recorded the largest monthly revenue decline among women customers at approximately Rs. 112 K.
Suggested actions: Review the performance and visibility of Set and Kurta products on Myntra. Compare pricing and promotions with competitors, review customer feedback, and test a targeted re-engagement campaign for previous women customers if customer-level data is available

Priority 2: Investigate and re-engage women customers on Amazon

Amazon recorded the second-largest monthly decline among women customers at approximately Rs.89K.
Suggested actions: Review Set and Kurta product listings, pricing, customer reviews, sponsored placements, and other factors that may have contributed to the decline. Test targeted marketing based on the findings.

Priority 3: Investigate why revenue from women customers declined

The analysis identifies where the decline occurred, but the available data does not fully explain why.
Suggested actions: Review customer feedback and returns, compare competitor pricing, and conduct customer research where possible. This should help determine whether the decline was related to product selection, pricing, competition, customer experience, or another factor.

Priority 4: Review the “Others” channel

The “Others” channel was the only channel to record revenue growth between March and November.
Suggested actions: Identify which sales channels are included in this group and investigate what contributed to the increase. Relevant findings could then be tested on larger marketplaces where appropriate.

Revenue Recovery Scenario

Revenue from women customers declined by approximately Rs.324,684 per month between March and November.
If Vrinda recovered 50% of the total monthly decline among women customers, the calculation would be:
Rs.324,684 × 50% × 12 = approximately Rs.1.95M per year

This can reasonably be presented as approximately Rs.2M in potential annual revenue recovery. This is a scenario estimate, not a forecast or guaranteed outcome. Actual results would depend on the causes of the decline and the effectiveness of any actions taken.

Methodology and SQL Code

The analysis was conducted in MySQL 8.0 using a dataset of 31,047 orders from January to December 2022. Revenue analysis was limited to orders with a Delivered status to exclude cancelled, refunded, and returned orders.

The six SQL queries below follow the same sequence used in the investigation.

Query 1 - Monthly Trend

```
SELECT
MONTH(Date) AS month_number,
MONTHNAME(Date) AS month_name,
COUNT(*) AS total_orders,
SUM(Amount) AS revenue,
ROUND(AVG(Amount), 2) AS avg_order_value
FROM vrindastoredata
WHERE Status = 'Delivered'
GROUP BY MONTH(Date), MONTHNAME(Date)
ORDER BY month_number;
```

Query 2 - Channel performance, March vs November

```
WITH channel_months AS ( SELECT
Channel,
SUM(CASE WHEN MONTH(Date) = 3 THEN Amount ELSE 0 END) AS march_revenue,
SUM(CASE WHEN MONTH(Date) = 11 THEN Amount ELSE 0 END) AS november_revenue
 FROM vrindastoredata
 WHERE Status = 'Delivered'
 GROUP BY Channel)
SELECT
Channel, march_revenue, november_revenue,
november_revenue - march_revenue AS revenue_change,
ROUND(((november_revenue - march_revenue) * 100.0 / march_revenue), 2) AS pct_change
FROM channel_months
ORDER BY pct_change;
```

Query 3 - Category performance, March vs November

```
WITH category_months AS
(SELECT Category,
SUM(CASE WHEN MONTH(Date) = 3 THEN Amount ELSE 0 END) AS march_revenue,
SUM(CASE WHEN MONTH(Date) = 11 THEN Amount ELSE 0 END) AS november_revenue
FROM vrindastoredata
WHERE Status = 'Delivered'
GROUP BY Category)
SELECT
Category, march_revenue, november_revenue,
november_revenue - march_revenue AS revenue_change,
ROUND(((november_revenue - march_revenue) * 100.0 / march_revenue), 2) AS pct_change
FROM category_months
ORDER BY revenue_change;
```

Query 4 - Top categories per channel (CTE + window function)

```
WITH ranked_categories AS
(SELECTChannel, Category, SUM(Amount) AS revenue,
RANK() OVER (PARTITION BY Channel ORDER BY SUM(Amount) DESC) AS category_rank
FROM vrindastoredata
WHERE Status = 'Delivered'
GROUP BY Channel, Category)
SELECT * FROM ranked_categories
WHERE category_rank <= 3
ORDER BY Channel, category_rank;
```

Query 5 - Demographic segmentation, March vs November

```
WITH gender_months AS
 (SELECT Gender, AgeGroup,
SUM(CASE WHEN MONTH(Date) = 3 THEN Amount ELSE 0 END) AS march_revenue,
SUM(CASE WHEN MONTH(Date) = 11 THEN Amount ELSE 0 END) AS november_revenue
FROM vrindastoredata
WHERE Status = 'Delivered'
GROUP BY Gender, AgeGroup)
SELECT
Gender, AgeGroup, march_revenue, november_revenue,
november_revenue - march_revenue AS revenue_change,
ROUND(((november_revenue - march_revenue) * 100.0 / march_revenue), 2) AS pct_change
FROM gender_months
ORDER BY pct_change;
```

Query 6 - Stress test: women customers by channel

```
WITH women_channel_march_nov AS
(SELECT Channel,
SUM(CASE WHEN MONTH(Date) = 3 THEN Amount ELSE 0 END) AS march_revenue,
SUM(CASE WHEN MONTH(Date) = 11 THEN Amount ELSE 0 END) AS november_revenue
FROM vrindastoredata
WHERE Status = 'Delivered'
AND Gender = 'Women'
GROUP BY Channel)
SELECT
Channel, march_revenue, november_revenue,
november_revenue - march_revenue AS revenue_change,
ROUND(((november_revenue - march_revenue) * 100.0 / march_revenue), 2) AS pct_change
FROM women_channel_march_nov
ORDER BY revenue_change;
```